LAYOUT - top view, front of the car to the RIGHT, driver side UP (R140 / R153). Not to scale; lengths in mm (estimates until measured at the car).
Each segment: length | wires in it | DR-25 heat-shrink size for its bundle (est. from the wire ODs). Lengths per wire (path + 100 mm at each end) are in the pinout table.

^ DRIVER SIDE (LH) at the top / PASSENGER SIDE (RH) at the bottom v

FRONT →
LHSILL LH sill, rear (under rear seat)
JB1 J/B No.1 + IE1 kick
LFP LH fender pass-through: FL sensor
WIPER Wiper motor F9
CHIAT Charge-pipe IAT
VSS Gearbox VSS
REVSW Reverse switch (TBD spot)
BOOST Boost solenoid
C_LHC LH front corner
R11 LH rear lamp R11
WSRL RL wheel-speed sensor
RLJ LH quarter: rear-light splices
COLUMN Steering column: cruise stalk
FLEX Flex fuel
BP_ENG_B3 FP3 branch point (B)
TURBO Turbo speed
ECT ECT
BP_ENG_A3 FP3 branch point (A)
CLUSTER Centre cluster (old meter spot)
VALET Valet switch (driver carpet)
PEDAL Accelerator pedal
CLNTP Coolant pressure
ETB ETB
JUMP Jump post
INJ4 Injector 4
COIL4 COP 4
TANK Fuel tank: pump + level sender
PI RealDash Pi (centre dash)
BP_COL pedal / column branch point
OILP Oil pressure
INJ3 Injector 3
STARTER_B Starter B+ stud
BP_ENG_B2 FP2 branch point (B)
COIL3 COP 3
FUEL Trunk LH: IX-FUEL inline
ALARM Anti-theft unit
FUELP Fuel pressure
INJ2 Injector 2
STARTER Starter S post
COIL2 COP 2
DASH Mid-dash: WS inlines, A/C switch
BP_DASH dash branch point
INJ1 Injector 1
CANJ CAN joins at the ECU
COIL1 COP 1
RPANEL Rear panel, ground BL
SIREN Alarm siren
ENG_BLOCK Engine block ground stud
C_FRONT Front centre: horns, return joins
HATCH High-mount stop H6
IAT IAT (manifold)
ECU ECU plugs A / B
BP_ENG_B1 FP1 branch point (B)
CSB3 Centre console: CSB3, status LED
EVAP Evaporator thermistor
BP_ENG_A2 FP2 branch point (A)
KNOCK Knock 1
SP ECU splice point
MAP MAP
FANR Radiator fan
CAM Cam sensor
R12 RH rear lamp R12
BP_ENG_A1 FP1 branch point (A)
ACAMP A/C amplifier taps
BP_AC A/C branch point
BH Bulkhead (DRB102, RH firewall)
BH_ENG Bulkhead (DRB102, engine half)
WSRR RR wheel-speed sensor
ALT_B Alternator B+ stud
FANC Condenser fan
ALT Alternator plug (IG / L / S)
BP_GB glove-box branch point
OILT Oil temp
BP_ENG_B0 FP0 branch point (B)
MRS MR-S EHPS pump
GB Glove box: POWER + LOGIC boxes, NEG bar
RL_ENG RADLOK pair (engine side)
ACC A/C compressor + pressure switch
LAMJ CAN-Lambda splices
FW_STR RADLOK pair (cabin side) + starter S pass
CRANK Crank sensor
BATT Trunk RH: battery, kill sw, relay block
RHSILL RH sill, rear
RHKICK RH kick panel (old R/B4, ground IG)
LAMBDA CAN-Lambda
RFP RH fender pass-through: FR sensor
RFG RH fender grommet (factory)
C_FEND RH inner fender (harness C)
C_RHC RH front corner

SEGMENTS

FROM	TO	MM	WIRES	BUNDLE	DR-25
ECU	SP	110	110	30.2	DR-25-1-1/2-0
SP	BP_AC	110	110	26.7	DR-25-1-1/2-0
BP_AC	BP_GB	110	112	26.9	DR-25-1-1/2-0
BP_GB	BH	100	64	18.3	DR-25-1-0
BP_GB	GB	150	60	21.7	DR-25-1-0
GB	RHKICK	250	16	13.1	DR-25-3/4-0
RHKICK	RHSILL	1600	8	6.1	DR-25-3/8-0
RHSILL	FUEL	400	3	5.3	DR-25-1/4-0
BP_AC	ACAMP	400	6	4.8	DR-25-1/4-0
BP_AC	EVAP	450	0	-	-
ECU	BP_DASH	30	60	27.8	DR-25-1-1/2-0
BP_DASH	DASH	300	19	23.5	DR-25-1-1/2-0
BP_DASH	BP_COL	220	48	25.0	DR-25-1-1/2-0
BP_COL	PEDAL	640	10	15.6	DR-25-3/4-0
BP_COL	JB1	650	34	20.5	DR-25-1-0
BP_DASH	ALARM	150	12	6.5	DR-25-3/8-0
ECU	CSB3	370	18	11.0	DR-25-1/2-0
ECU	CANJ	50	17	16.5	DR-25-1-0
BH_ENG	BP_ENG_A1	100	41	24.5	DR-25-1-1/2-0
BP_ENG_A1	CRANK	1410	3	7.9	DR-25-3/8-0
BP_ENG_A1	MRS	1360	7	7.8	DR-25-3/8-0
BP_ENG_A1	BP_ENG_A2	200	39	24.0	DR-25-1-1/2-0
BP_ENG_A2	MAP	130	3	3.3	DR-25-3/16-0
BP_ENG_A2	IAT	400	2	2.7	DR-25-3/16-0
BP_ENG_A2	ETB	430	8	15.3	DR-25-3/4-0
BP_ENG_A2	FLEX	230	4	10.8	DR-25-1/2-0
BP_ENG_A2	CAM	850	4	8.1	DR-25-3/8-0
BP_ENG_A2	INJ4	160	8	6.3	DR-25-3/8-0
INJ4	INJ3	180	6	5.5	DR-25-1/4-0
INJ3	INJ2	380	4	4.5	DR-25-1/4-0
INJ2	INJ1	170	2	3.2	DR-25-3/16-0
BP_ENG_A2	COIL4	100	12	7.7	DR-25-3/8-0
COIL4	COIL3	160	9	6.7	DR-25-3/8-0
COIL3	COIL2	180	6	5.5	DR-25-1/4-0
COIL2	COIL1	180	3	3.9	DR-25-3/16-0
BP_ENG_A2	ENG_BLOCK	300	5	8.3	DR-25-1/2-0
BP_ENG_A2	BP_ENG_A3	450	8	11.4	DR-25-3/4-0
BP_ENG_A3	TURBO	150	4	10.8	DR-25-1/2-0
BP_ENG_A3	BOOST	280	2	2.7	DR-25-3/16-0
BP_ENG_A3	ECT	700	2	2.7	DR-25-3/16-0
BH_ENG	BP_ENG_B0	150	7	11.0	DR-25-1/2-0
BP_ENG_B0	LAMJ	90	5	10.6	DR-25-1/2-0
LAMJ	LAMBDA	150	4	3.8	DR-25-3/16-0
BP_ENG_B0	OILT	960	2	2.7	DR-25-3/16-0
BH_ENG	BP_ENG_B1	200	18	14.9	DR-25-3/4-0
BP_ENG_B1	KNOCK	1260	3	9.4	DR-25-1/2-0
BP_ENG_B1	BP_ENG_B2	200	15	11.7	DR-25-3/4-0
BP_ENG_B2	FUELP	30	3	3.3	DR-25-3/16-0
BP_ENG_B2	OILP	580	3	3.3	DR-25-3/16-0
BP_ENG_B2	CLNTP	1020	3	3.3	DR-25-3/16-0
BP_ENG_B2	BP_ENG_B3	300	8	11.7	DR-25-3/4-0
BP_ENG_B3	CHIAT	360	2	2.7	DR-25-3/16-0
BP_ENG_B3	VSS	680	4	10.8	DR-25-1/2-0
BP_ENG_B3	REVSW	880	2	3.6	DR-25-3/16-0
GB	FW_STR	400	5	6.0	DR-25-3/8-0
FW_STR	STARTER	2470	1	4.6	DR-25-1/4-0
CANJ	CLUSTER	400	11	11.0	DR-25-1/2-0
CANJ	PI	250	6	9.0	DR-25-1/2-0
BP_COL	COLUMN	400	5	4.4	DR-25-1/4-0
ALARM	VALET	250	2	2.7	DR-25-3/16-0
GB	RFG	300	29	16.4	DR-25-3/4-0
RFG	C_FEND	600	26	16.2	DR-25-3/4-0
C_FEND	ACC	250	4	4.3	DR-25-1/4-0
C_FEND	MRS	400	4	6.9	DR-25-3/8-0
C_FEND	C_RHC	500	20	14.3	DR-25-3/4-0
C_RHC	FANC	300	4	12.2	DR-25-3/4-0
FANC	FANR	300	2	8.6	DR-25-1/2-0
C_RHC	C_FRONT	700	19	11.5	DR-25-3/4-0
C_FRONT	C_LHC	700	11	8.8	DR-25-1/2-0
RFG	SIREN	500	2	2.7	DR-25-3/16-0
RFG	ALT	400	3	3.7	DR-25-3/16-0
RHKICK	RFP	300	2	7.6	DR-25-3/8-0
JB1	LFP	300	6	9.6	DR-25-1/2-0
LFP	WIPER	300	4	5.8	DR-25-3/8-0
JB1	LHSILL	1800	5	11.0	DR-25-1/2-0
LHSILL	RLJ	800	1	1.9	DR-25-3/16-0
RLJ	R11	300	0	-	-
RLJ	RPANEL	600	1	1.9	DR-25-3/16-0
RPANEL	R12	600	0	-	-
RPANEL	HATCH	400	0	-	-
LHSILL	WSRL	400	2	7.6	DR-25-3/8-0
LHSILL	WSRR	1300	2	7.6	DR-25-3/8-0
FUEL	TANK	300	3	5.3	DR-25-1/4-0
RHSILL	BATT	600	7	7.1	DR-25-3/8-0
RL_ENG	ENG_BLOCK	690	3	3.3	DR-25-3/16-0
RL_ENG	STARTER_B	1500	2	2.7	DR-25-3/16-0
ALT_B	RL_ENG	300	1	1.9	DR-25-3/16-0
STARTER_B	JUMP	210	1	1.9	DR-25-3/16-0

NODES

ECU ECU plugs A / B (centre dash, factory ECM spot)
SP ECU splice point (bussed DT plugs + soldered splices)
BP_AC A/C branch
BP_GB glove-box branch
BH DRB102 bulkhead, cabin half (RH firewall)
GB Glove box: POWER + LOGIC boxes (every cabin fuse and relay), NEG bar, SW12 bus, blower lead
RHKICK RH kick panel (old R/B No.4 spot): ST1 / heater taps, ground IG
RHSILL RH sill, rear end (the run turns across the trunk)
FUEL Trunk LH: IX-FUEL inline to the fuel sender
ACAMP A/C amplifier (A18) taps
EVAP Evaporator case: thermistor A24 leads
BP_DASH dash branch
DASH Mid-dash above the ECU: front + rear WS inlines, A/C switch lead
BP_COL pedal / column branch
PEDAL Accelerator pedal (driver footwell)
JB1 J/B No.1 + cowl plug IE1, LH kick panel: IG2 / tail / turn taps, door switch lead, ground ID (Routes 2, 7, 15)
ALARM Anti-theft unit (centre dash, left of the ECU)
CSB3 Centre console: CSB3 under the centre cluster, its switch inputs and opto board; alarm status LED (Routes 12, 14)
CANJ CAN trunk at the ECU: CAN H / L joins, ECU 120 ohm end (Route 8)
BH_ENG DRB102 bulkhead, engine half (RH firewall)
BP_ENG_A1 A trunk branch point FP1 (just off the bulkhead, passenger side; Route 3 point 3)
BP_ENG_A2 A trunk branch point FP2 (top of the head, centre; Route 3 point 4)
BP_ENG_A3 A trunk branch point FP3 (driver side of the head; Route 3 point 5)
CRANK Crank VR sensor (Dan's 'crank below' spot; Route 3 point 6)
MRS MR-S EHPS pump plugs B / C (pump at the old ABS block spot; Route 3 point 7)
MAP MAP sensor (Route 3 point 9)
IAT IAT sensor, intake manifold (Route 3 point 8)
ETB Bosch 74.5 mm ETB (Route 3 point 10)
FLEX Flex fuel sensor (driver side, top; Route 3 point 12)
CAM Cam Hall sensor (centre of the head, behind the COP row; Route 3 point 11)
INJ4 Injector 4 (injector row enters at cyl 4; Route 3 point 13)
INJ3 Injector 3
INJ2 Injector 2
INJ1 Injector 1
COIL4 COP 4 (coil row enters at cyl 4; Route 3 point 14)
COIL3 COP 3
COIL2 COP 2
COIL1 COP 1
ENG_BLOCK Engine block ground stud (RADLOK engine ground H6; Route 1 point 5)
TURBO Turbo speed sensor (Route 3 point 17)
BOOST Boost control solenoid (Route 3 point 16)
ECT ECT sensor (Route 3 point 15)
BP_ENG_B0 B trunk branch point FP0 (passenger leg; Route 4 point 3)
LAMJ CAN-Lambda joins: CAN H / L, +12 V and ground splices (240 mm from the bulkhead)
LAMBDA Link CAN-Lambda controller (wideband O2; Route 4 point 8)
OILT Oil temp sender (Route 4 point 7)
BP_ENG_B1 B trunk branch point FP1 (rear of block, centre; Route 4 point 4)
KNOCK Knock sensor 1 (factory K1 leader; Route 4 point 9)
BP_ENG_B2 B trunk branch point FP2 (rear of block, driver side; Route 4 point 5)
FUELP Fuel pressure sensor (Route 4 point 10)
OILP Oil pressure sensor (Route 4 point 12)
CLNTP Coolant pressure sensor (rear of block, driver side; Route 4 point 11)
BP_ENG_B3 B trunk branch point FP3 (driver side, low; Route 4 point 6)
CHIAT Charge-pipe IAT sensor (Route 4 point 13)
VSS Gearbox VSS (spot estimated in Route 4, point 14)
REVSW Back-up (reverse) switch B1 on the transaxle - spot TBD (not on Route 4)
FW_STR RH firewall: RADLOK pair, cabin side, and the starter S wire pass - spot TBD (Route 1 point 1)
STARTER Starter solenoid S (50) post (starter: lower-middle of the engine; Route 1 point 2)
CLUSTER Centre cluster (old meter spot): LED board on the meter plugs C11 / C12, cluster CAN board + buck, J/B No....
PI RealDash Pi + USB-CAN adapter, centre dash, and the dash 12 V split (Route 8)
COLUMN Steering column: cruise stalk C14 flying leads (Route 12, spot approximate)
VALET Valet / program switch lead, under the carpet to the driver-side carpet centre, service coil (Route 14)
RFG Existing factory RH fender grommet, RH front inner fender corner: harness C, alarm siren + hood pin, alternator IG ...
RFP RH fender pass-through by the strut tower: FR wheel-speed plug (Route 9 rev 2)
LFP LH fender pass-through by the strut tower, driver kick: FL wheel-speed plug; wiper feed out to the cowl (Routes 2, 9)
WIPER Front wiper motor F9, under the cowl, driver side (Route 2 point 18)
LHSILL LH sill, rear end: under the rear seat (rear wheel-speed legs meet; rear-light run turns to the quarter panel; Rout...
RLJ LH quarter panel centre (OEM joint BP1): rear-light splices (Route 15)
R11 LH rear lamp R11 (socket pigtails)
RPANEL Rear panel centre: licence lamp L1, ground point BL, trunk pin switch (Routes 14, 15)
HATCH High-mount stop light H6 (hatch spoiler)
R12 RH rear lamp R12 (socket pigtails)
WSRL RL ABS sensor plug (rear left; Route 10)
WSRR RR ABS sensor plug (rear right; its cable crosses under the rear seat to the LH sill, Route 10)
TANK Fuel tank: fuel pump and the level sender (DT 2-way), trunk LH ground stud (Route 11 rev 2)
BATT Trunk RH rear corner: battery, kill switch, battery relay block (FP 40 A fuse + fuel pump relay K-FP) (Route 1 point...
C_FEND Harness C: down the RH inner fender (A/C branch, EPS pump) (Route 2 rev 12)
ACC A/C compressor A4 clutch lead + dual pressure switch A5 (Route 2 point 7)
C_RHC RH front corner: headlamp H2, turn / clearance F8, retract motor R3, washer W1, hood pin, ground EA (Route 2 ...
FANC Condenser fan (own connector; first stop on the fan branch, Route 2 point 9)
FANR Radiator fan (own connector; end of the fan branch, Route 2 point 10)
C_FRONT Core support, front centre: horn split, both horns (DT 2-way each), headlamp / retract / clearance return joins ...
C_LHC LH front corner: headlamp H1, turn / clearance F7, retract motor R2, ground EB (Route 2 point 16)
SIREN Alarm siren: centre of the firewall behind the engine, leads through the RH fender grommet (Route 14)
ALT Alternator G2 round 3-pin plug (IG / L / S, through the RH fender grommet) - rear RH of the engine under the bulkhe...
RL_ENG RADLOK + / - pair, engine side (RH firewall) (Route 1 point 1)
ALT_B Alternator B+ stud, rear RH of the engine (Route 1 point 3)
STARTER_B Starter B+ stud (H4, H5, H8 lugs) (Route 1 point 2)
JUMP Engine-bay jump post (Route 1 point 6, proposed spot)

The ECU splice point (SP) is ONE place by the ECU plugs (Dan 2026-10-19): +5 V on BUS-5V + BUS-5V-2, Gnd Out A24 on BUS-GNDOUT-A, B22 on BUS-GNDOUT-B (bussed DEUTSCH DT, DECISIONS R169 parts), +8 V, switched 12 V, the A5 feed and the ECU power ground soldered there. The ECU ground runs alone to the POWER box NEG bar (R216).

REVISIONS

REV	DATE	BY	DESCRIPTION
B	2026-10-25	DG	Surgical rewire written into the master: 21 decided circuits, D16 A/C, OEM starter / reverse / rear l...
C	2026-10-26	DG	A15 lock-sensor input takes a modified tacho: Ign 8 through 4.7 k· to pin 9 (N61 workaround).
D	2026-10-26	DG	Car-check answers: if the ignition-switch black-orange wire is at the engine plug, remove it; A/C s...
E	2026-10-26	DG	C11 is hardware starter interlock only; Aux 8 spare. Oil LED stays on CSB3 L4.

ST185-harness - every wiring harness of the car, one drawing set
Layout - top view, segment lengths, wire counts, DR-25 sizes

DRAWING NO.	REV	SHEET	SIZE	DRAWN	DATE
ST185-harness-LAY	E	1 OF 1	1647 x 950 mm	DG	2026-10-26

ST185 Celica (living-relation) - generated by docs/harness/tools/build_drawings.py from docs/harness/MASTER-wirelist.csv; not to scale